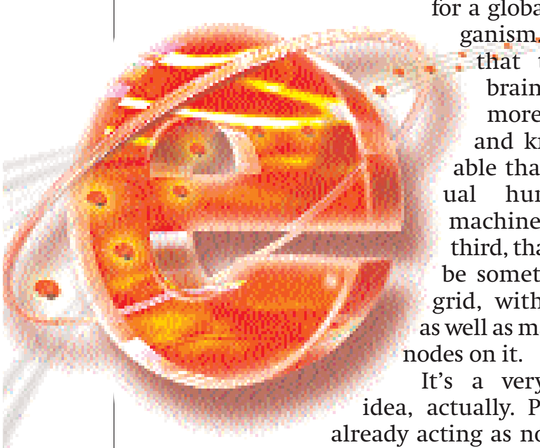
its power—the utility of the network increases exponentially as more users are added. Third, both the brain and the Internet have their information protocols.

In an information-driven world, then, think of many intelligent units connected together, with the Internet as the medium. The information is the thought process. The individual nodes—whether mechanical or human—are the neurons. It's a huge collective brain, no less. But it's a brain in its infancy: it doesn't think for itself, it doesn't learn, it doesn't solve problems... What several theoreticians have proposed is that the next logical step in the development of the Internet is of it becoming a Global Brain or Overbrain.

On the Internet, each node—an applet, site, server, or whatever —is dumb in itself, but the Net as a whole can exhibit intelligent behaviour. The concept of an Overbrain is based on a few broad ideas: first, that inter-human communication, being as advanced as it is now, is sufficient to make for a global superorganism. Second, that the Overbrain will be more powerful and knowledgeable than individual humans or machines. And third, that it would be something of a grid, with humans as well as machines as nodes on it.



It's a very natural idea, actually. People are already acting as nodes. Take Google Answers, for example. You pay and get your query answered—it doesn't matter to you whether your query has been answered by a bot or a human. The expert who answers your question is a node. You play a game on the Web with a stranger. You're both nodes. You make a Wikipedia entry. You're a node.

In what follows, we introduce what people have been calling the Global Brain, which we call the Overbrain. We look at mechanisms by which the Internet could become more intelligent, and ask what could happen thereafter.

Similar Sites, Similar People

Today's Internet has the trappings of a brain, but not those of a particularly smart one. Think about Amazon's recommendation system. When you buy a book or CD, you're prompted with something like "People who bought this item also bought..." This could happen all over the Net! But as of now, when you visit a Web page, you aren't presented with something like "People who visited this page also visited..."

There could also be links from a page you visited to pages that "similar" people also visited. Web sites such as del.icio.us introduce the concept of "similar people."

These concepts—of similar sites and similar people—are important. Treating both sites and people as nodes, we're talking here about clustering. Many aver that in a biological brain, it is clusters of neurons that make up ideas and concepts. Fragmented sets of information points just don't cut it: there needs to be organisation. In practical terms, an organised Web would be much more useful than the fragmented collection it is today. And in theoretical terms, an Overbrain would need to be organised if it is to learn and think.

> **The Web will learn from—and adapt to—the behaviour and needs of its users. No human knowledge "injected" into the Web need ever be lost any more**

A Learning Web

The human brain implements what is known as "associative learning." Neurons that are activated one after the other have the connection strength between them increased. Similarly, concepts that are more frequently used together become more closely interlinked. For example, if you frequently think about "mouse" and "keyboard" at the same time, the two become more tightly integrated together—possibly under a new heading, such as "input devices."

Applying this to the Web would mean increasing the linkage between two pages depending on how close they are in terms of usage. So if a user views page A and then page B, the linkage between them can be increased, or a new link can be placed between them. If the user views page C after that, there will also be a link from A to C. Now, these links will need a minimum strength in order to be displayed, and so some links could die out altogether. Of course, there's the issue of unrelated sites being visited in succession, but that's addressed by the fact that the links between such sites will ultimately die out.

An example of this is at the Global Brain Group (GBG), associated with the Principia Cybernetica Project (PCP). J Bollen, working with F Heylighen, chair of the GBG and AI researcher at the Free University of Brussels, has put up a smart server that does just this. The result is "a dynamic system of strengthening and weakening links between different pages."

Bollen continues, "These ever-shifting hyperlinks bear a remarkable resemblance to connections that grow and fade in a human brain. On the Principia Cybernetica Web (the Web site of the PCP; http://pespmc1.vub.ac.be), algorithms will reinforce popular links... It's the first step on the road to the global brain."

The important thing about this is that the Web is learning from—and adapting to—the behaviour and needs of its users. No human knowledge "injected" into the Web need ever be lost any more. Distant documents could get linked together. Clusters of ideas could form. Take the example of gorillas and conservation; one site could be that of Greenpeace, which mentions gorillas in passing. The other site could be about the social behaviour of gorillas. Mere keyword matching would not link these two as similar; if the Web could learn, it would eventually tie them together.

What if the Web were intelligent enough to learn concepts on its own? The Overbrain could do data mining, where interesting patterns are observed from a mass of data and presented to the user. If a sufficient number of people who consume a certain kind of food (purchased on the Web, of course) develop a vitamin deficiency, the Overbrain could mark this up as a syndrome, and warn buyers of that food about the deficiency.